COMP4442

SERVICE AND CLOUD COMPUTING

Cloud Computing Platforms
Part-1

Computing Platforms

In IT, a platform is any hardware or software used to host an application or service.

You can leverage a platform to accomplish something in a simpler, faster, and better way

e.g., a desktop platform consists of hardware, an OS and coordinating programs that use the instruction set for a particular processor or microprocessor, which creates a foundation that ensures object code will execute successfully

Cloud computing hosts different combinations of hardware and software platforms on its own infrastructure and offer them as a service

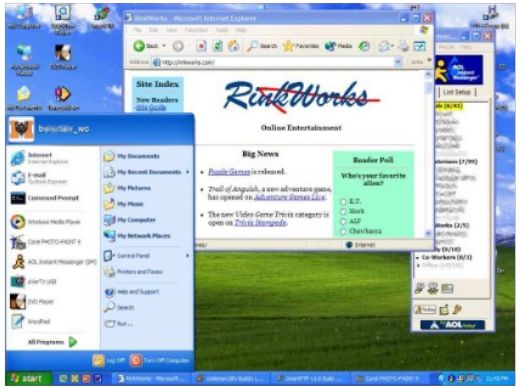
PaaS : software stack built upon the architecture and provide support for software development with higher level abstractions

Provide low-level functionality ready-made: Users only need to take care of data and application

Developers can leverage pre-existing code rather than starting from scratch and writing everything.

PC

- CPU – Intel, AMD
- OS – Microsoft Windows



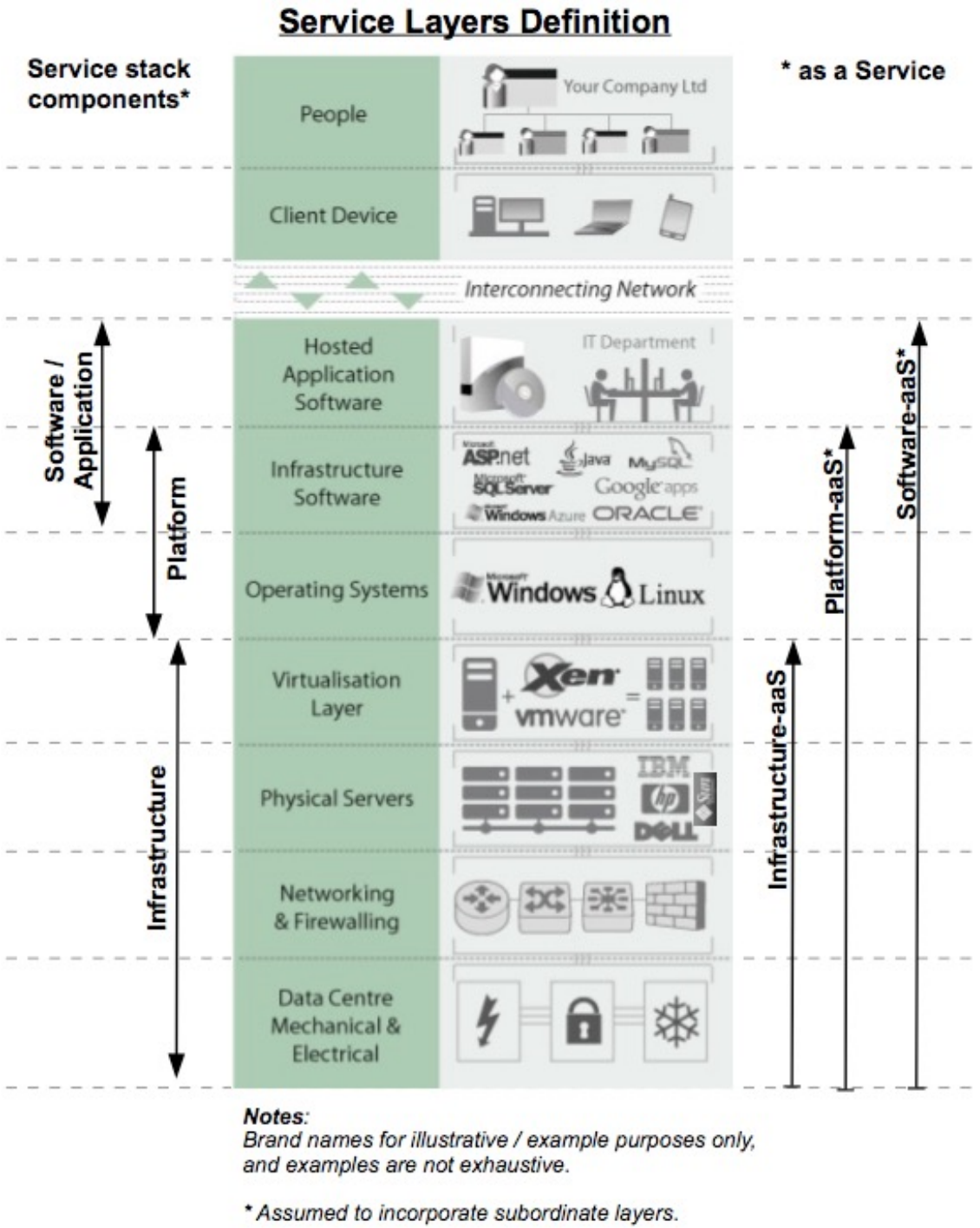
Mac

- CPU – Motorola
- OS – Apple Mac OS

PaaS Revisited

PaaS is a category of cloud service that provides a computing platform and environment to allow develops to build applications and services

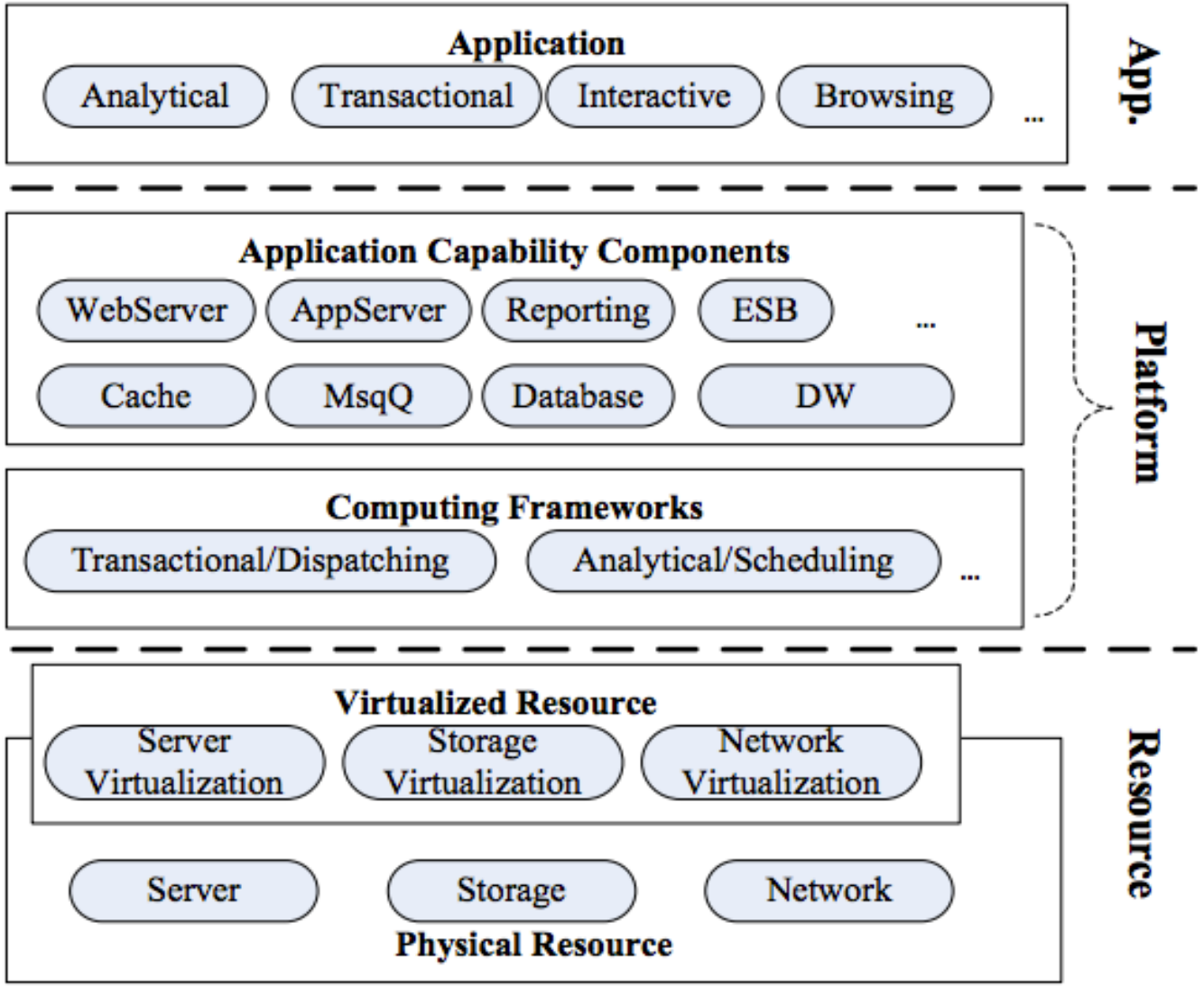
- Abstraction between cloud applications and the IaaS
- Functionality: support efficient and cost-effective process for creation and deployment of high-quality applications
  - Development automation
  - Infrastructure management
  - Computing framework managing task scheduling etc.
  - Container orchestration
  - …
- Fundamentally provide elastic scaling of cloud applications



PaaS Functionality

PaaS enables us to do the following:

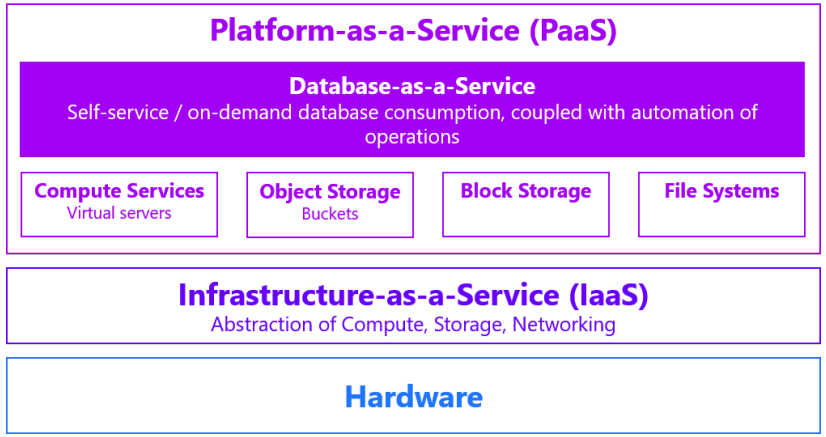
- Eliminates the complexity of selecting, purchasing, configuring, and managing hardware and software by leveraging key middleware services
- Access a complete stack of development and deployment tools via a web browse, and overcome the challenges of managing lots of individual tools by providing a suite of integrated and standardized tools – OS, security products, …
- Enjoy fully isolated environment on top of a multi-tenant architecture
- Easier and Agile application deployment
  - Simplifies prototyping and deploying startup solutions

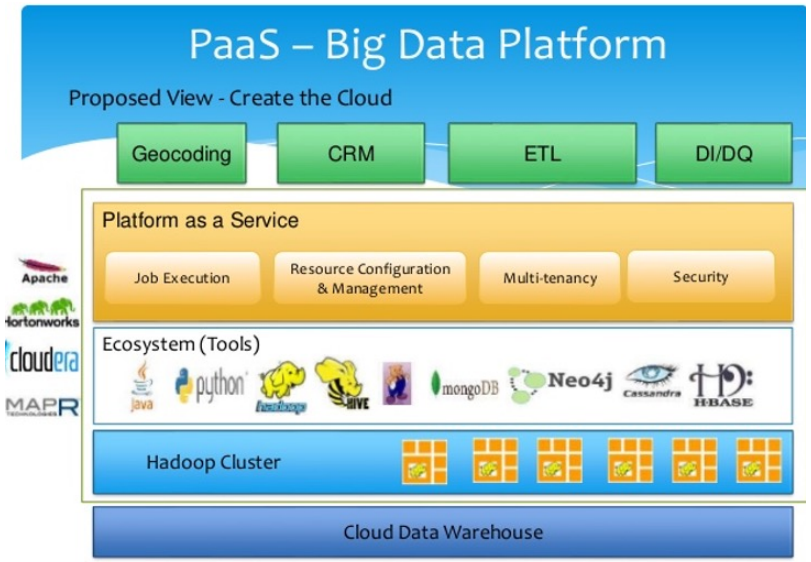


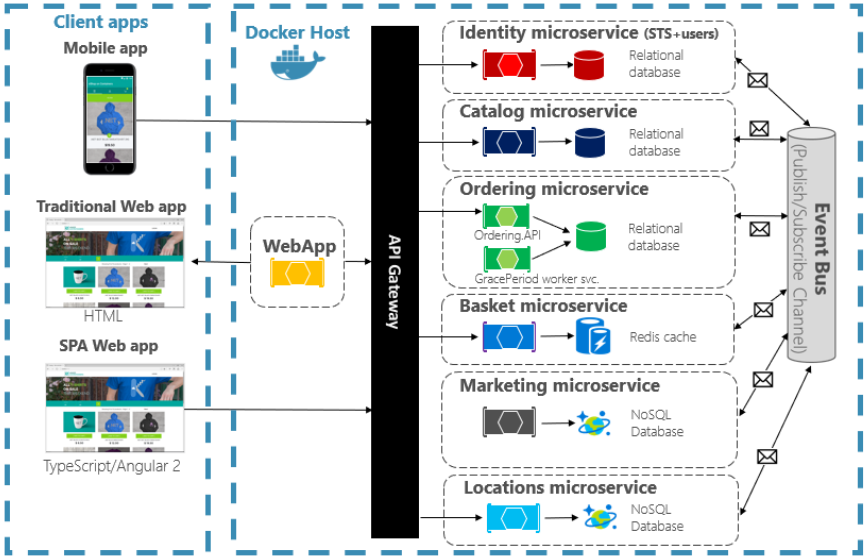
PaaS Examples

Example PaaS services:

- Middleware (e.g., messaging & queueing)
- Application runtime and management (e.g., load balancers, deployment, …)
- Databases (DynamoDB)
- Development platform (Hadoop, Spark, …)







PaaS Example: Message Queues*

Eliminate complexity and overhead associated with managing and operating messaging software & infrastructure

Message queues middleware provides an asynchronous communication channel for applications to interact

- Sender and receiver of the message do not need to be ready for communication at the same time.

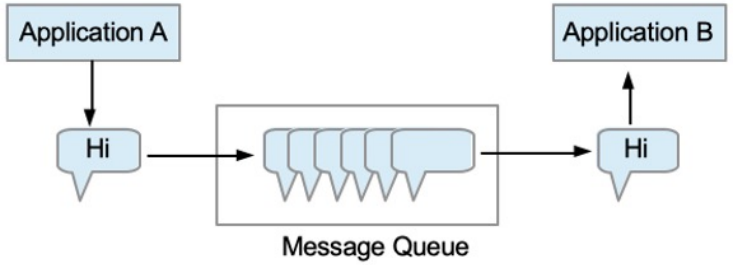
Needed in many scenarios:

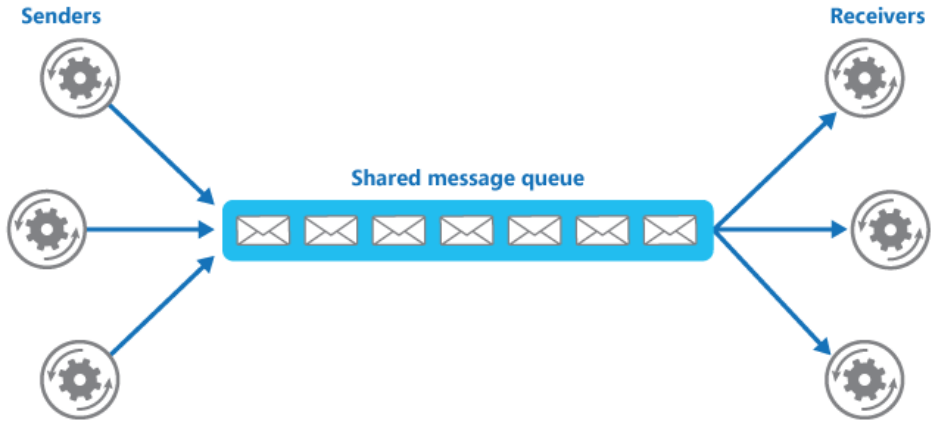
- application take a time to generate a response
- non time sensitive jobs process data in the background
- distribute data processing among many servers

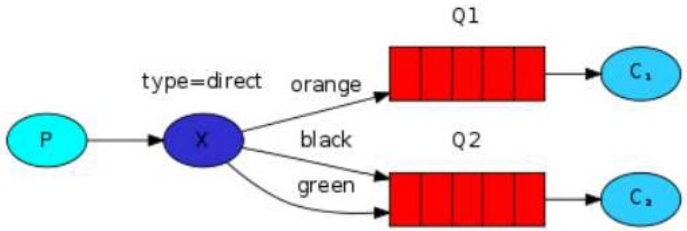
A message broker can be used to mediate interactions amongst applications, and decouple the applications

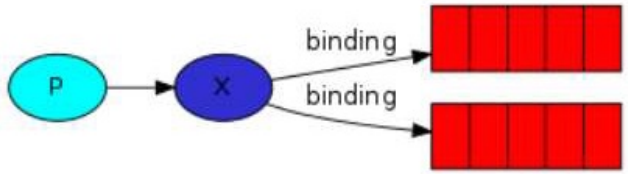
Routing types

- Direct vs. Fan-out





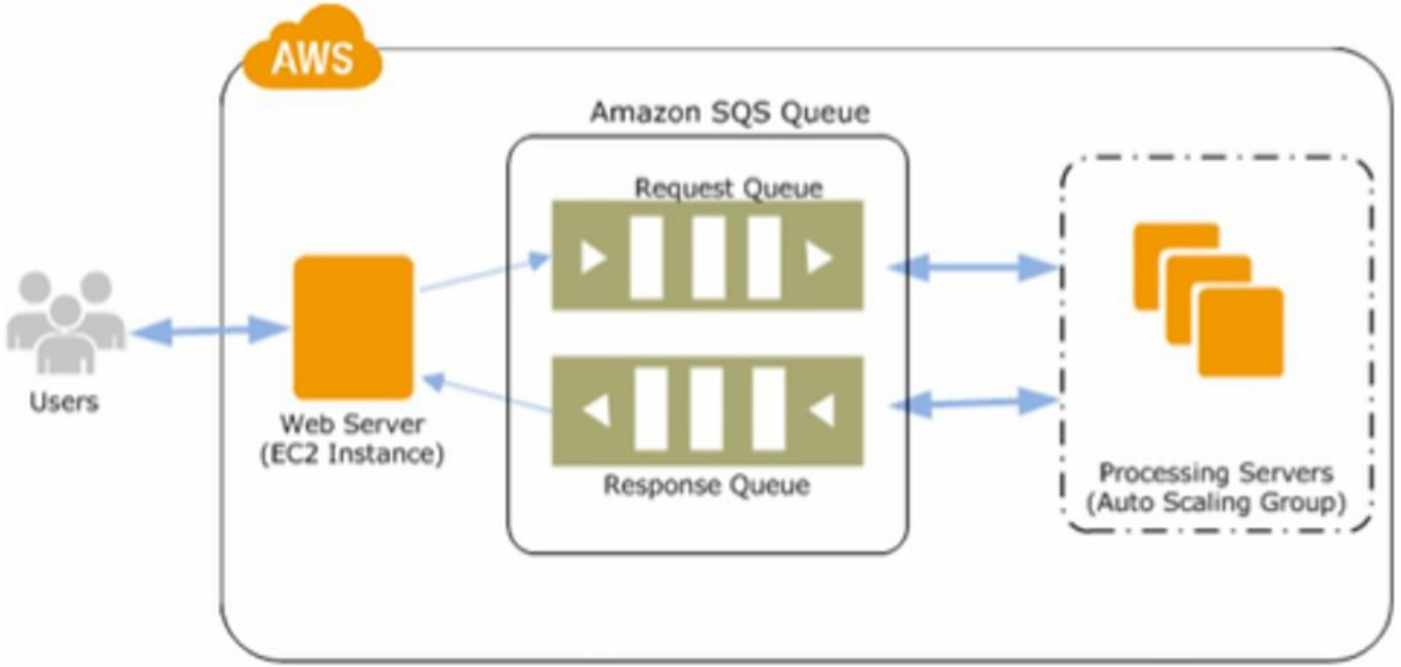




PaaS Example: AWS SQS*

AWS SQS (Simple Queue Service)

- A fully managed message queuing service that let users send, store, and receive messages between software components at any volume of data.
- Pull based system offering
  - high throughput
  - Best-effort ordering
  - At-least-once delivery by default
  - Messages can be sent in batch and Long pulling can reduces extraneous polling
- Can get status & alerts (queue depth, message rate, ..) via CloudWatch
- Two types of queues



create-queue,
delete-queue
send-message
receive-message
…

Standard Queue	FIFO Queues
Unlimited number of transactions per second	First-in-first-out delivery, limited to 300 transactions per second
Guarantee that a message is delivered at least once	Exactly-once processing
Ex: Schedule multiple entries to be added to a database.	Ex: Prevent a student from enrolling in a course before registering for an account.

PaaS Example: AWS SNS*

AWS SNS (Simple Notification Service)

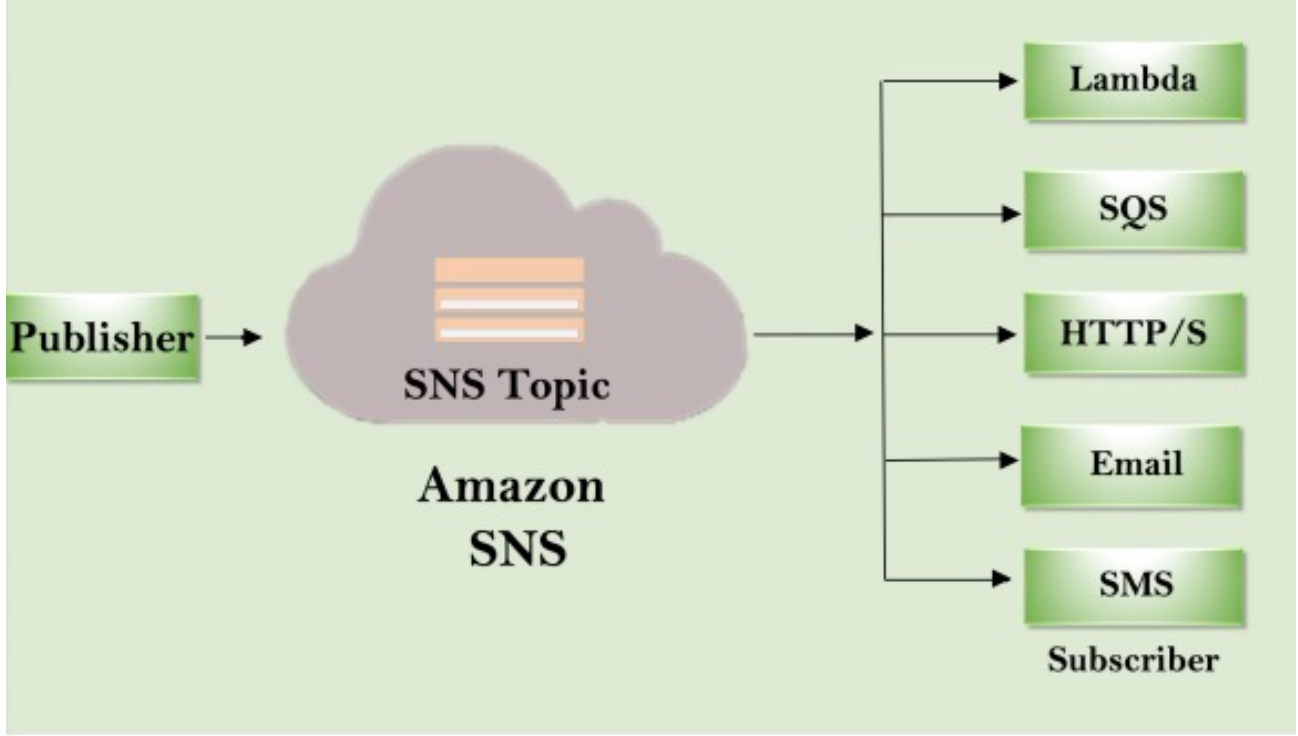
A fast, flexible, fully managed notification service that lets you send individual messages or to fan-out messages to large numbers of recipients.

Push based systems: send push notifications to mobile device users, email recipients, or even send messages to other distributed services.

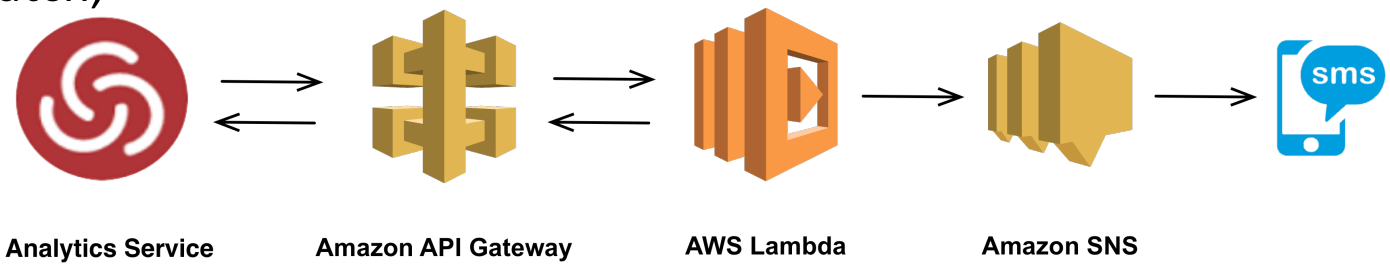
Deliver messages instantly to applications/users and eliminate polling in your apps

SDK available in iOS, Android, Java, Python, PHP, Node.js or .NET.

Can get delivery status information via CloudWatch, e.g. success and failure rates.



CreateTopic, Subscribe, and Publish



PaaS Example: AWS SQS / SNS*

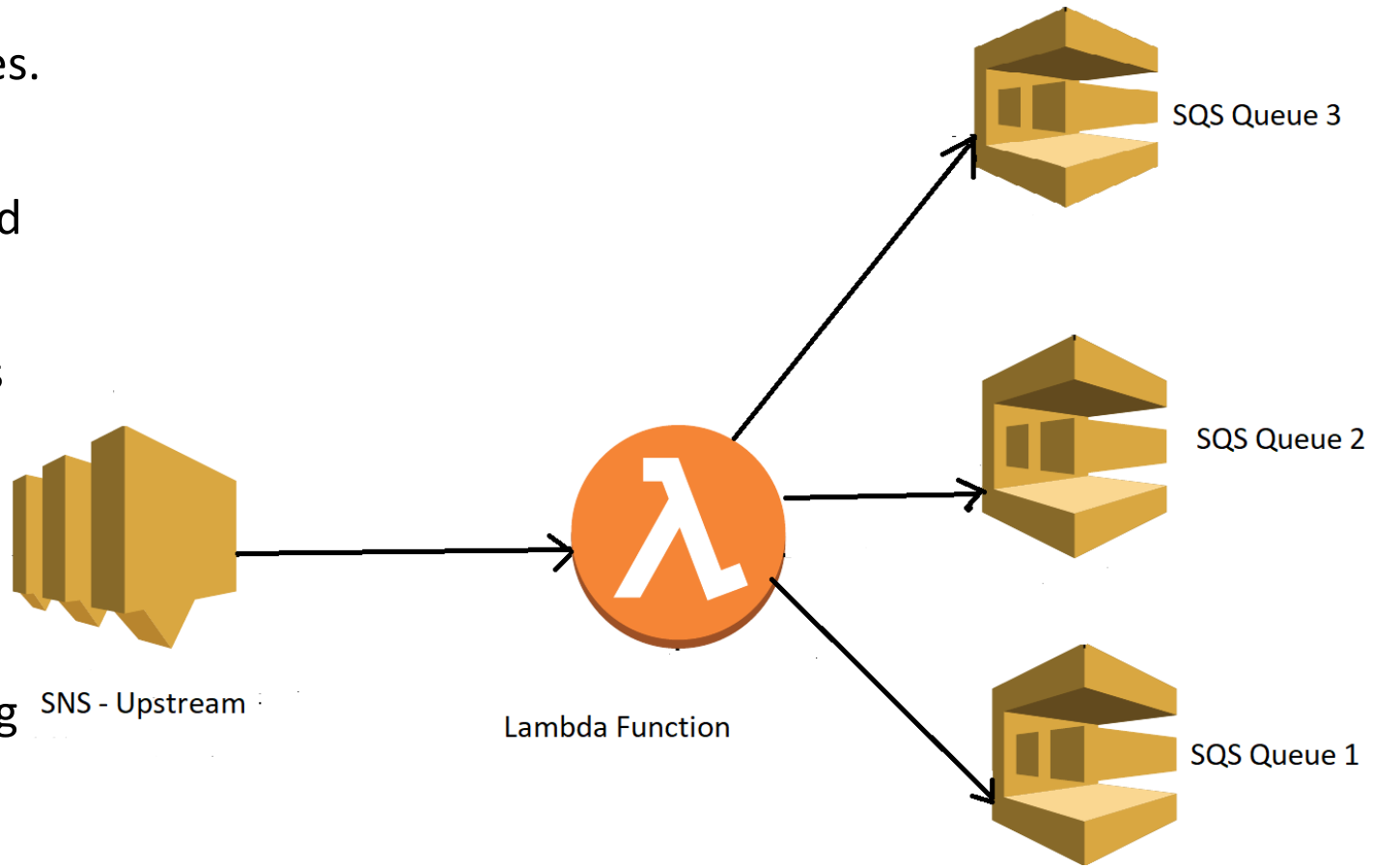
When combined with SNS, developers can 'fanout' identical messages to multiple standard SQS queues.

With fanout, a message published to an SNS topic is distributed to a number of SQS standard queues in parallel.

Using fanout, developers can build applications that take advantage of parallel, asynchronous processing.

For example, developers could publish a message to a topic every time a new image is uploaded. Independent processes, each reading from a separate queue, could generate thumbnails, perform image recognition, and store metadata about the image.

SNS does not currently support FIFO ordering, and as such fanout to FIFO queues is not supported

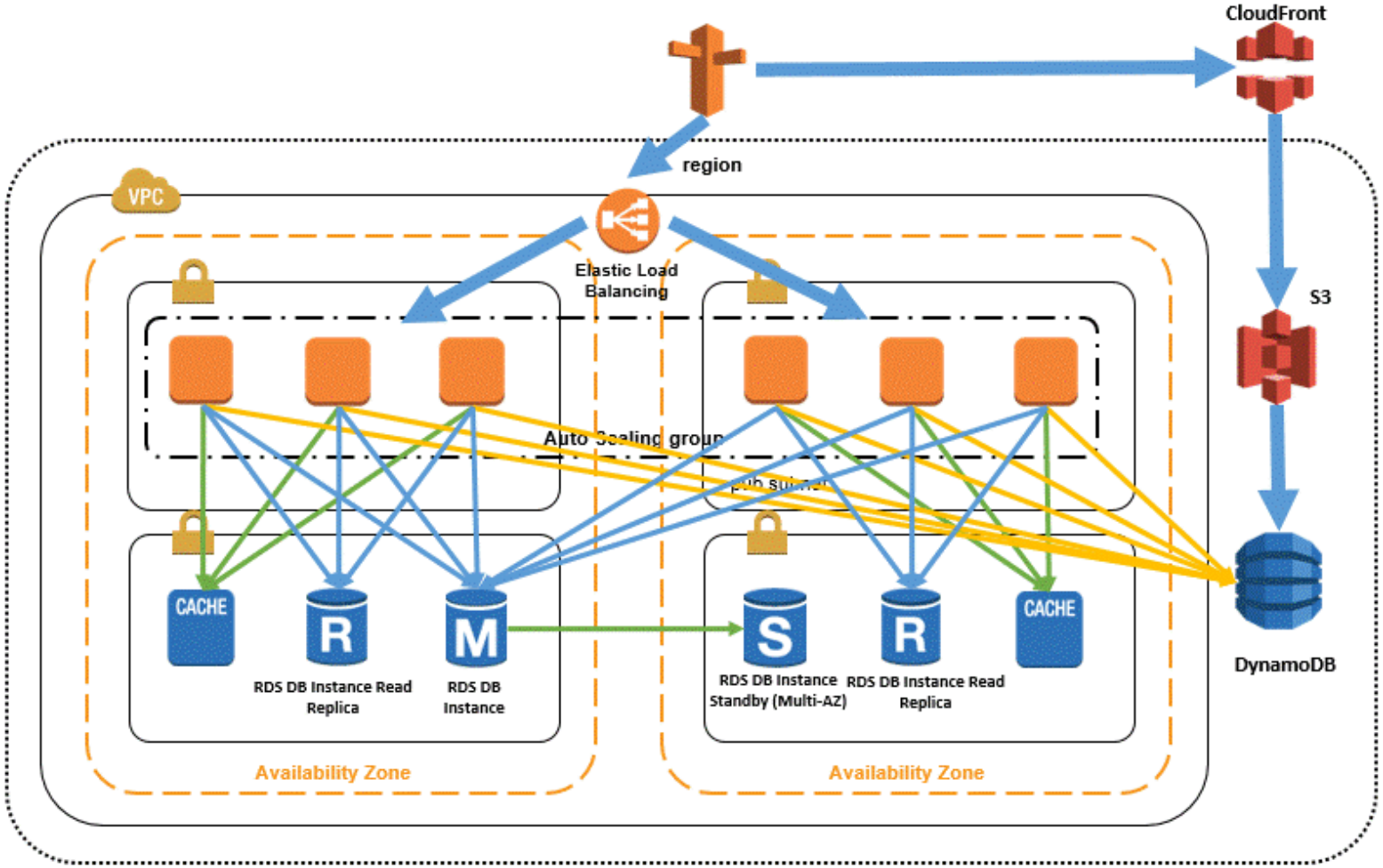


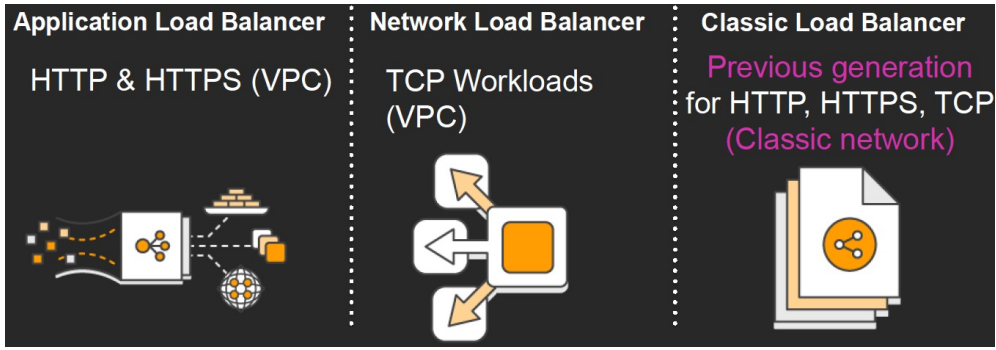
PaaS Example: AWS Elastic Load Balancer

Automatically scales and distributes incoming application traffic to multiple targets, such as EC2 instances, containers and IP addresses;

Monitors the health of targets, and routes traffic only to the healthy targets.

Span across multiple availability zones within a single region

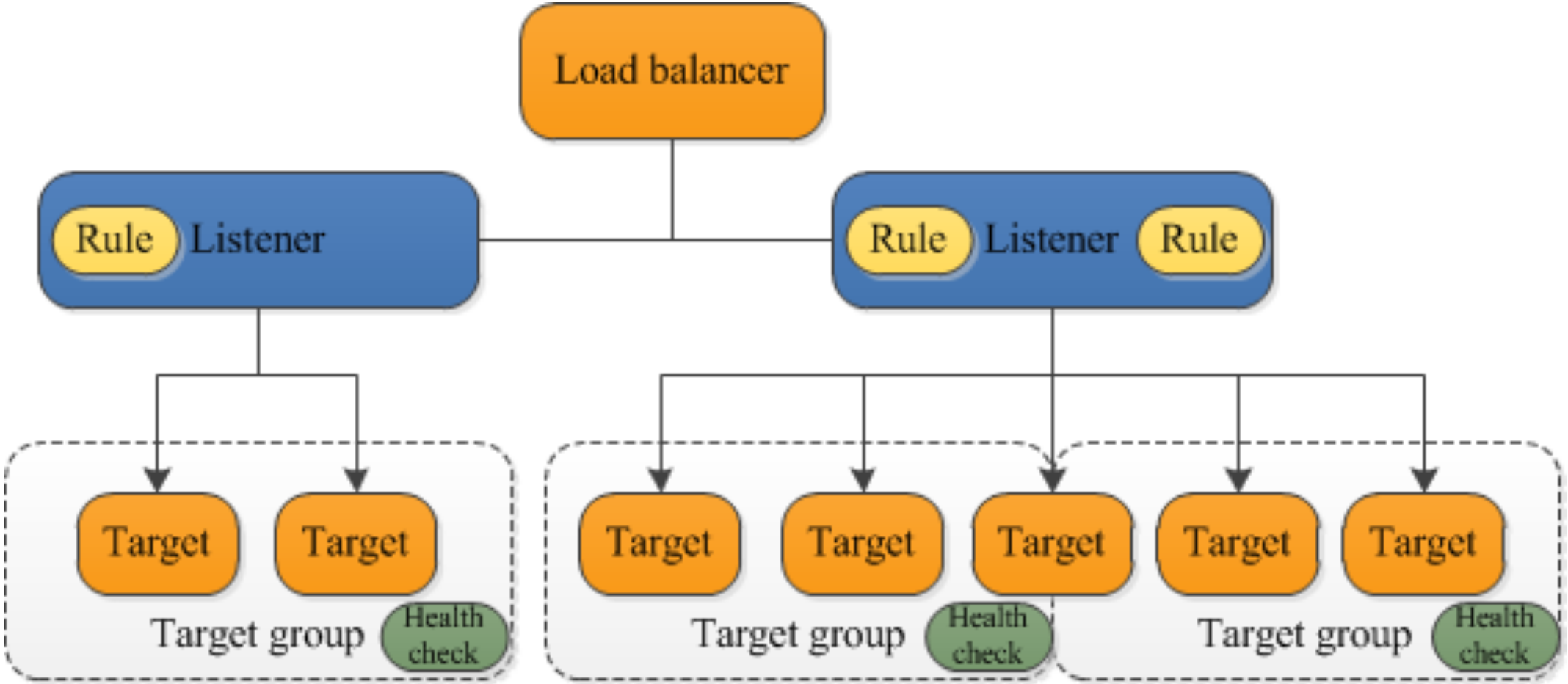




PaaS Example: AWS Elastic Load Balancer

Application Load Balancer

- layer 7 load-balanced platform: allows requests to be routed to different applications behind a single load balancer
- Content-based routing: route based on path or host field in the HTTP header
- Support for microservices and container-based applications



PaaS Example: AWS Elastic Beanstalk

AWS Elastic Beanstalk

An easy-to-use service for deploying and scaling web applications and services developed using Java, .NET, PHP, Node.js, Python, Ruby, Go and Docker on familiar servers such as Apache, IIS, …

Simply upload your code and Elastic Beanstalk does the provisioning automatically - from capacity provisioning, load balancing and automatic scaling to status monitoring of the application.

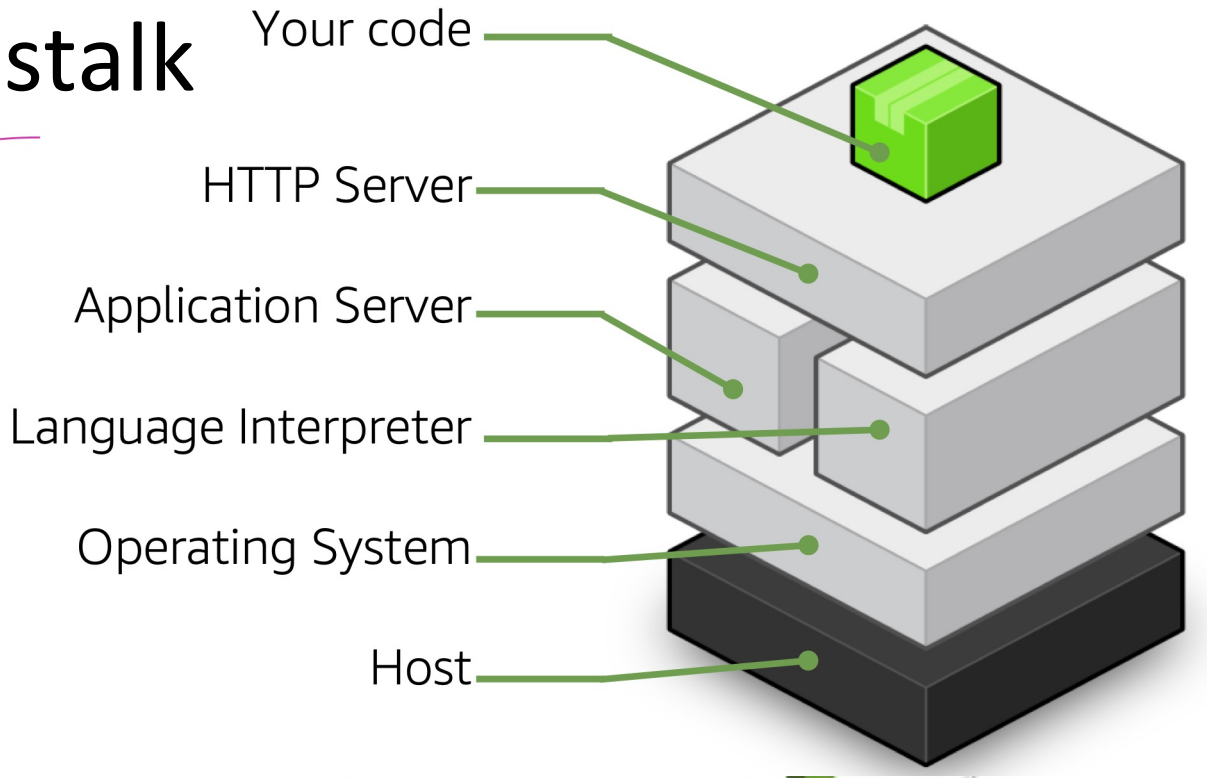
Meanwhile, gives you complete control over the AWS resources behind your application and gives you access to the underlying resources at any time.

E.g., running web applications on AWS

AWS has multiple services that enable you to run web application: compute, storage, database, network, …

While each service is powerful and simple to use, running a web application requires that you learn multiple services and leaves the selection, configuration, and management up to you

PaaS Example: AWS Elastic Beanstalk

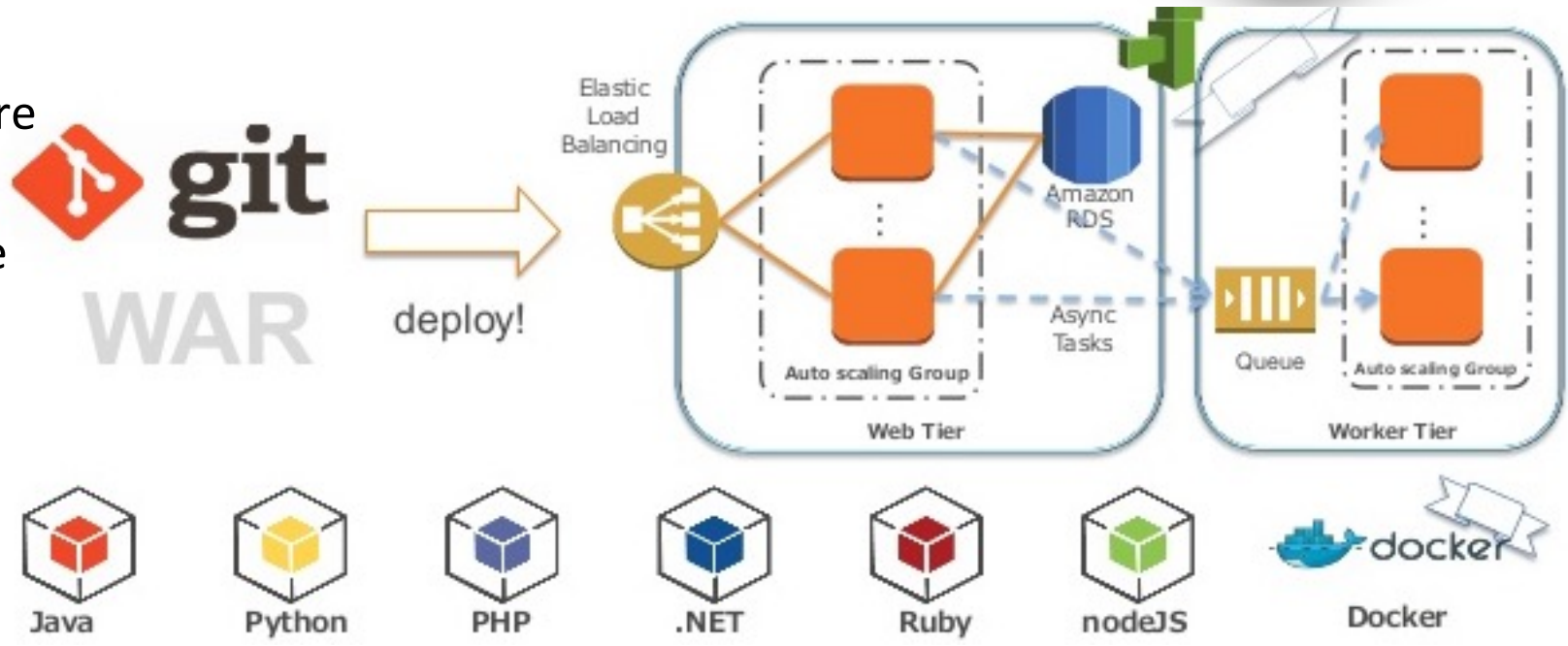


AWS Elastic Beanstalk provides the easiest way to get your web applications up and running in minutes!

Elastic Beans task is a managed service that automatically handles things like Infrastructure provisioning and configuration, Deployment, Load balancing, Auto scaling, Health monitoring, Analysis and debugging, …

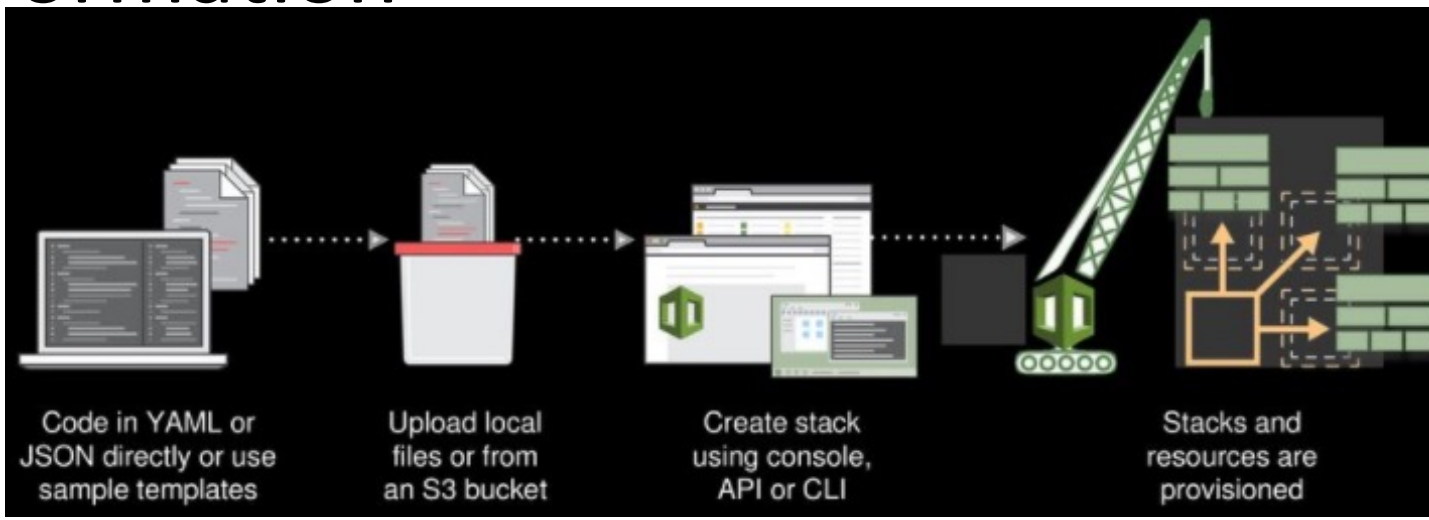
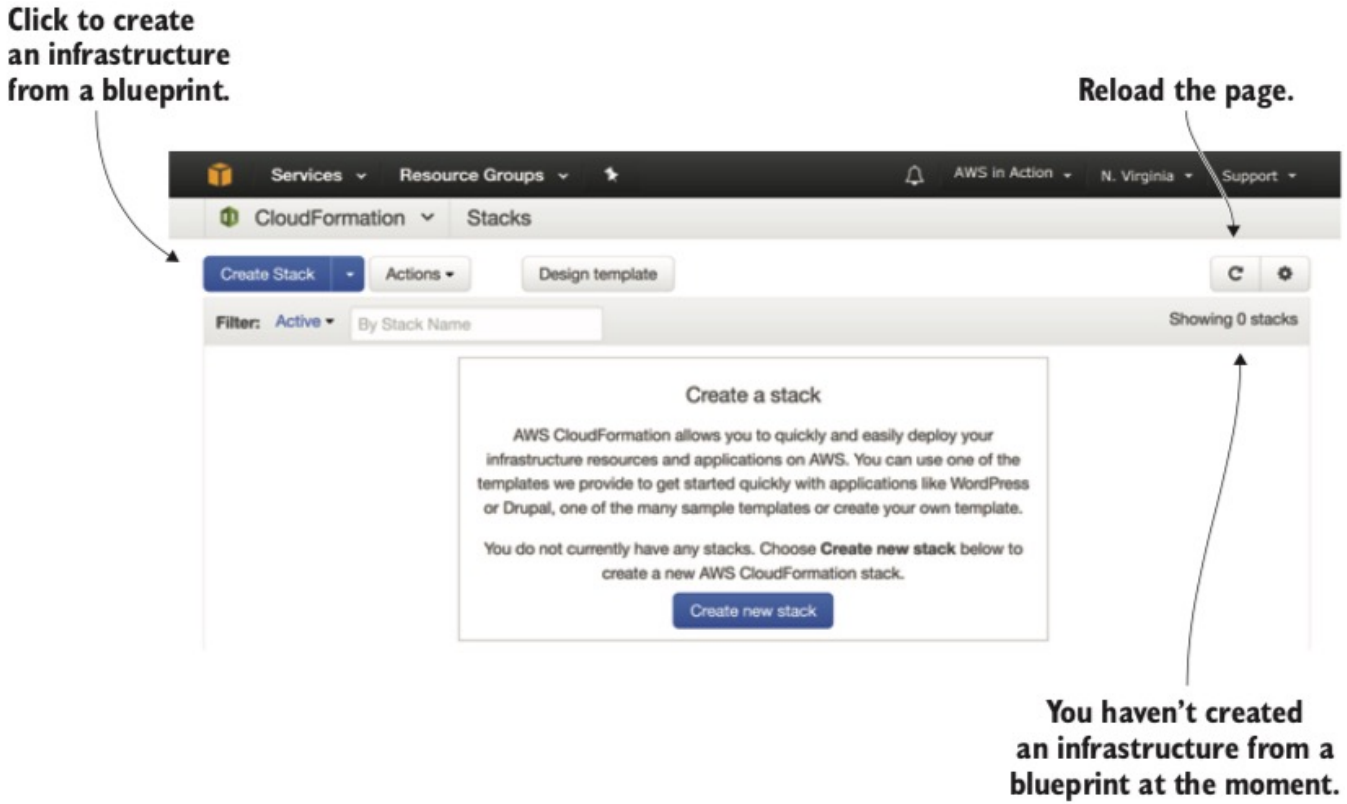
What Elastic Beanstalk do:

- Provisioned the necessary infrastructure resources
- Bundled source code together with the corresponding platform (OS, runtime, application server, and web server)
- Safely deployed your web application



Elastic Beanstalk vs. CloudFormation

An alternative for you to automate the application setup and deployment is to use AWS CloudFormation

- A way to “form the cloud”: uses *templates* to make the process easier
- Automatically do the following in background:
  - Create a load balancer (ELB)
  - Create a MySQL database (RDS)
  - Create a network filesystem (EFS)
  - Create and attach firewall rules (security groups)
  - Create VMs running web servers
- Provides access to infrastructure components and allow you to manage and control the application stacks and resources needed for your applications

CloudFormation belongs to ”Infrastructure Build Tools”, but not under PaaS as Beanstalk